

JEE MAIN REVISION MASTER: CIRCLE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Equation of Circle, Center, Radius & Tangents	3 PYQs	Conic Sections

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Equation of Circle, Center, Radius & Tangents	$\text{Center} = (-g, -f)$, $r = \sqrt{g^2 + f^2 - c}$, $\text{Condition for Tangency: } c = \pm a\sqrt{1+m^2}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Equation of Circle, Center, Radius & Tangents (Moderate Yield — 3 PYQs)

Core Principle: General equation of circle: $x^2 + y^2 + 2gx + 2fy + c = 0$. Center is $(-g, -f)$ and radius is $r = \sqrt{g^2 + f^2 - c}$.
Tangent at (x_1, y_1) : $xx_1 + yy_1 + g(x+x_1) + f(y+y_1) + c = 0$. Condition for line $y = mx + c$ to be tangent to $x^2 + y^2 = a^2$: $c^2 = a^2(1+m^2)$. Condition for two circles to cut orthogonally: $2g_1g_2 + 2f_1f_2 = c_1 + c_2$.

■ **Caution / JEE Pitfall:** Before calculating center $(-g, -f)$, ensure coefficients of x^2 and y^2 are reduced to 1!

■ **Speed Trick:** Length of tangent from external point (x_1, y_1) to circle $S=0$ is simply $L = \sqrt{S_1}$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The area enclosed by the curve $x=\sqrt{3}\cos\theta, y=\sqrt{3}\sin\theta$ is..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "If in two circles, arcs of the same length subtend angles 30° and 78° at the centre, then the ratio of their radii is..." [Ans: N/A]
- [2018 Shift] Q#3 (MCQ): "The maximum area of a rectangle inscribed in the circle $(x+1)^2 + (y-3)^2 = 64$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The area enclosed by the curve $x=\sqrt{3}\cos\theta, y=\sqrt{3}\sin\theta$ is

- (A) $\sqrt{3}\pi$ sq. units
- (B) 9π sq. units
- (C) 6π sq. units
- (D) 3π sq. units

Correct Answer: None

Solution Analysis: Given, $x=\sqrt{3}\cos\theta, y=\sqrt{3}\sin\theta$ We know the standard parametric form of a circle is $x=a\cos\theta, y=a\sin\theta$. This represents a circle with centre at the origin and radius a . Here, $a=\sqrt{3}$. So the radius of the circle is $\sqrt{3}$. Area enclosed by a circle is πr^2 . So, $\text{Area}=\pi(\sqrt{3})^2$. $\text{Area}=3\pi$. Hence, the area enclosed by the curve is 3π sq. units. Correct option: Option D.